Regularization under a limited training budget

Abstract

Controlled exploratory Fashion-MNIST experiments compare five regularization treatments across three seeds. Results below are measured CPU experiments, not a full-data benchmark.

1 Methods

The same small CNN, seed-specific initialization, fixed train/validation/test indices and eight-epoch budget were used for all treatments. Training used 3,000 images, validation and test used 750 each. AdamW learning rate was 0.001; dropout was 0.3 and weight decay 0.01 where enabled. Checkpoints were selected by validation loss. Training accuracy was evaluated without augmentation and dropout.

2 Results

Treatment	Mean test accuracy	Sample SD
baseline	0.7507	0.0083
dropout	0.7471	0.0198
weight_decay	0.7493	0.0069
augmentation	0.7222	0.0113
combined	0.7209	0.0235

3 Discussion and limitations

Three seeds and a small test subset do not establish population-level significance. A smaller generalization gap can reflect underfitting. This is an exploratory extension following a short pilot; it does not reproduce published protocols. Full records and the complete literature review are in the accompanying Markdown report.

References

[1] Srivastava et al. Dropout. JMLR 15, 2014.

[2] Loshchilov and Hutter. Decoupled Weight Decay Regularization. ICLR, 2019.

[3] Simard et al. Best Practices for Convolutional Neural Networks Applied to Visual Document Analysis. ICDAR, 2003.

[4] Zhang et al. Understanding deep learning requires rethinking generalization. ICLR, 2017.

[5] Xiao et al. Fashion-MNIST. arXiv:1708.07747, 2017.